

Pathogenesis of type 2 diabetes risk in black Africans: a South African perspective

■ J. H. Goedecke¹  & T. Olsson² 

From the ¹Non-Communicable Diseases Research Unit, South African Medical Research Council, Cape Town, South Africa; and ²Department of Public Health and Clinical Medicine, Umeå University, Umeå, Sweden

Content List – Read more articles from the symposium: “Obesity and Type 2 Diabetes: Understanding the Role of Ethnicity”.

Abstract. Goedecke JH, Olsson T (South African Medical Research Council, Cape Town, South Africa; and Umeå University Hospital, Umeå, Sweden). Pathogenesis of type 2 diabetes risk in black Africans: a South African perspective (Review Symposium). *J Intern Med*; 2020; **288**: 284–294.

The prevalence of type 2 diabetes (T2D) is higher in black Africans than their European counterparts. This review summarizes the research exploring the pathogenesis of T2D in populations of African ancestry compared to white Europeans and shows that the pathogenesis differs by ethnicity. Black Africans present with a phenotype of low insulin sensitivity and hyperinsulinaemia as a result of increased insulin secretion and reduced hepatic insulin clearance. Whether hyperinsulinaemia precedes insulin resistance or is merely a compensatory mechanism is yet to be determined. Black

Africans have lower visceral adipose tissue and ectopic fat deposition and greater peripheral (gluteo-femoral) fat deposition than their European counterparts. This suggests that black Africans are more sensitive to the effects of ectopic fat deposition, or alternatively, that ectopic fat is not an important mediator of T2D in black Africans. Importantly, ethnic disparities in T2D risk factors may be confounded by differences in sociocultural and lifestyle factors. Future longitudinal and dietary intervention studies, in combination with genetic analyses, are needed for a better understanding of the pathophysiology of T2D in black Africans. This will be key for effective prevention and management strategies.

Keywords: hyperinsulinaemia, insulin resistance, hepatic insulin clearance, ectopic fat, African.

Introduction

Whilst the estimated prevalence of type 2 diabetes (T2D) in sub-Saharan Africa (SSA) is relatively low (4.2%), the projected rate of increase over the next 25 years exceeds that of other regions, with the prevalence in SSA expected to increase by 2.5-fold by 2045 [1]. The impact of T2D in SSA is reflected by the high estimated T2D-associated deaths (~312 000 deaths in 2017), with 73% of these being in people under the age of 60 years, a proportion higher than any other region in the world [1]. Within SSA, South Africa has the second highest number of people with T2D and has an estimated national prevalence of 5.1% [1]. Notably, T2D was the 2nd leading cause of death in South Africa in 2016 (5.5% of deaths), but the highest

amongst women (7.2% of deaths), with the greatest number of deaths recorded in the black South African population [2]. A study in urban-dwelling black South Africans showed higher T2D prevalence rates in women compared to men (13.1% vs. 11.3%), which has increased significantly over the past 20 years [3]. Although the pathogenesis of T2D has been widely studied in European populations, less is known in Africans, with very limited information in African men. An understanding of the pathogenesis of T2D is essential for effective prevention and management.

Ethnic differences in insulin sensitivity, secretion and clearance

The risk for T2D is dependent on the relationship between insulin sensitivity and secretion. This relationship is hyperbolic in nature [4] (Fig. 1), such that at low levels of insulin sensitivity, a small change in insulin sensitivity would result in a large

[Correction added on 27 May 2020, after first online publication: the article title has been corrected in this current version.]

insulin response, whereas at high levels of insulin sensitivity, a small change in insulin sensitivity would result in a small increase in insulin response. Notably, we have shown that at the same levels of body fat percentage and fasting glycaemia, both normal weight and obese black South African women have significantly lower insulin sensitivity than their European counterparts, but hyperscrete insulin to maintain normoglycaemia [5] (Fig. 1). These findings are consistent with other studies on black Africans from Africa [6] and the USA [7–9]. Furthermore, Kodama et al. [8] in a systematic review, including studies of Africans, Europeans and East Asians, reported significant ethnic differences in the stabilization points of insulin sensitivity and insulin response to maintain normal glucose concentrations. The hyperbolic relationship between insulin sensitivity (S_I) and the acute insulin response to glucose (AIRg) allows for the product to be calculated ($S_I \times \text{AIRg}$), known as the disposition index (DI), which provides an estimate of beta-cell function for a given level of insulin sensitivity [4]. Results from the Insulin Resistance Atherosclerosis study (IRAS) multi-ethnic cohort study showed that low DI, independently of ethnicity, glucose tolerance, family history of T2D and obesity predicted the incidence of T2D at 5-year follow-up [9]. Whilst some studies have shown no differences in DI between black and white women [5,10], suggesting an appropriate compensatory beta-cell response to the low levels of insulin sensitivity in black women, others have shown higher DI in black compared to white women [12], which is incongruent to their higher risk for T2D [13]. Rather, the higher beta-cell capacity and insulin response is associated with hyperinsulinaemia, which is characteristic of black African populations [5,10–12]. The hyperinsulinaemia may however be at least partly related to primary alterations in insulin clearance.

Insulin clearance is performed predominately by the liver (~40–80%), and to a lesser extent, other tissues such as skeletal muscle and kidney [14,15]. Several studies from West Africa, the UK and USA have reported reduced hepatic insulin clearance [7,12,16,17], but not extra-hepatic insulin clearance in black Africans compared to their European counterparts [17]. Notably, Osei et al. [16] showed both lower basal and postprandial insulin clearance in African Americans, Ghanaian immigrants to USA and native Ghanaians compared to European Americans. A reduction in hepatic insulin clearance was previously proposed

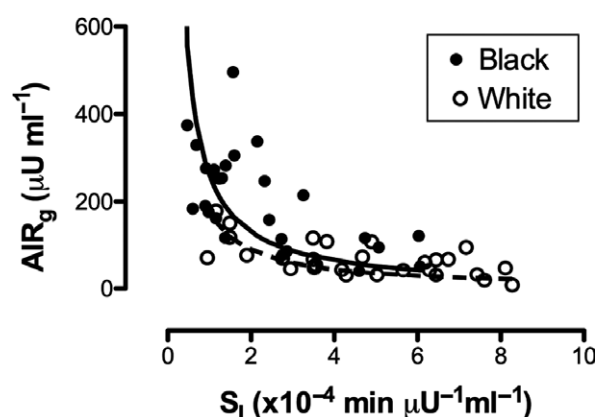


Fig. 1 The hyperbolic relationship between insulin sensitivity and insulin response in black and white South African women, adapted from Goedecke et al. [5]. S_I , insulin sensitivity; AIRg, acute insulin response to glucose. Solid represent the relationship in black women; broken line represents the relationship in white women.

to be a compensatory mechanism to preserve beta-cell function in the face of insulin resistance [18], based on experimental models. Of major interest is that a recent study in a multi-ethnic cohort of adolescents showed that lower hepatic (and not extra-hepatic) insulin clearance was an independent determinant of reduced beta-cell function at 2-year follow-up [19]. Further, results from the IRAS study showed that lower whole-body insulin clearance predicted the development of T2D in nondiabetic African Americans and Hispanics, independent of lifestyle factors, total and visceral adiposity and insulin secretion [20]. However, this association was not independent of insulin sensitivity [20]. Accordingly, insulin clearance has been purported to be an alternative explanation for the development of T2D in black Africans.

When exploring sex differences in insulin sensitivity and response between black South African men and women with normoglycaemia, we showed that the stabilization points of insulin sensitivity and response differed by sex [21]. Women had lower levels of insulin sensitivity than their male counterparts but increased their insulin response to maintain normoglycaemia. With increasing age, insulin sensitivity remained unchanged and the insulin response decreased similarly in men and women, with the largest declines observed in early adulthood [21]. These findings are in contrast to studies in largely European populations, where adult men have been shown to have lower insulin

sensitivity and being at greater risk for T2D than women [22]. Although the prevalence of overweight and obesity is significantly lower in black South African men compared to women (31% vs. 65%) [23], the increase in BMI in black South African men, albeit at a slower rate than women [24,25], poses a great risk for future T2D. However, there is a paucity of studies examining the determinants of T2D in black African men. This may be due to their lower perceived risk due to their lower BMI, but also due to their low participation rate in research trials [3,26]. Preliminary results from a cross-sectional study of middle-aged black South African men and women found that when adjusting for differences in BMI, men had lower basal insulin clearance and lower insulin response, and accordingly a lower DI, suggesting increased risk for T2D due to reduced beta-cell function in men compared to women (Kufe et al, unpublished). In agreement with this, a recent study from the UK comparing men of west African and European ancestry with early T2D showed that the west African men had lower insulin secretory function with concomitantly lower hepatic insulin clearance rate in response to oral glucose [27].

There is increasing evidence to suggest that hyperinsulinaemia *per se* may drive obesity, insulin resistance and T2D. The impact of hyperinsulinaemia, in combination with a high glycaemic index diet, on obesity risk in African Americans is eloquently reviewed by Barbara Gower in this symposium series (Gower in this issue). This is supported by our earlier work that showed that the frequency of the tetranucleotide insertion polymorphism in the proinsulin gene (IVS-69, rs3842740) that increases translation efficiency, and which is absent in non-Africans but present at a frequency as high as 25% in African populations, was higher in black South African women with extreme obesity [28]. Further we showed that obese women with the IVS-69 insertion allele had significantly greater fat mass, in particular abdominal subcutaneous adipose tissue (SAT) than wild-type homozygotes [28].

Hyperinsulinaemia may also directly drive insulin resistance [29–31] by causing receptor and postreceptor defects [32] and a defect in the nonoxidative glucose disposal (glycogen synthetic pathway) [31]. In a recent study in nondiabetic European adults, primary hypersecretion of insulin predicted deterioration in glucose tolerance [33]. Further, hypersecretion of insulin was associated with hepatic insulin resistance and mild increases in serum

triglycerides, liver enzymes and gluconeogenic substrates [33]. Based on these results, Trico et al. [33] proposed that the phenotype of insulin hypersecretors is characterized by the antecedence of hepatic insulin resistance over peripheral insulin resistance, leading to inappropriately increased endogenous glucose output and mild hyperglycaemia, coupled with increased cycling of gluconeogenic substrates that exacerbates the hyperglycaemia. However, this clinical phenotype differs to that shown in black Africans in whom hyperinsulinaemia is attributed to a combination of increased insulin secretion and reduced hepatic clearance [7,16], or in the case of west African men living in London, where hyperinsulinaemia seems predominantly due to reduced hepatic insulin clearance [34], and not merely increased insulin secretion as shown in the study by Trico et al. [33]. Although their findings were confirmed in a small cohort of bi-ethnic (African American and European) obese American adolescents, the African Americans had a lower prevalence of insulin hypersecretion than their European counterparts, which contradicts previous studies (as reported above). Notably, using a two-step euglycaemic, hyperinsulinaemic clamp in a small sample of obese black and white South African women matched for age and BMI [35], we showed that although basal insulin concentrations were higher in black than white South African women, black South African women had lower hepatic fat deposition and a correspondingly greater hepatic insulin sensitivity, characterized by greater suppression of endogenous glucose production, compared to their white counterparts [35]. Moreover, Chung et al. [36] showed that premenopausal African American women had less hepatic fat and importantly had lower rates of gluconeogenesis compared to their European counterparts matched for age, BMI and whole-body insulin sensitivity. Further, they showed that for any given level of whole-body insulin sensitivity, hepatic insulin sensitivity was higher in African American than European women [36].

In summary, the findings of these studies consistently show that irrespective of their geographical location, black Africans present with lower whole-body insulin sensitivity and hyperinsulinaemia as a result of increased insulin secretion and reduced hepatic insulin clearance compared to their European counterparts (Table 1). It is not clear whether the hyperinsulinaemia precedes insulin resistance or is merely a compensatory mechanism. Nonetheless, these findings suggest that the genetic

Table 1. Ethnic differences in insulin sensitivity, secretion and clearance

Compared to white Europeans, black Africans present with		References
↓ insulin whole-body insulin sensitivity		[5-12,16]
↑ hyperinsulinaemia		[5,7-12,16]
↑ insulin response/secretion		[5,7-12,16]
↓ hepatic insulin clearance		[7,12,16,17]
↔ extra-hepatic insulin clearance		[17]
↑ hepatic insulin sensitivity		[35,36]
↓ gluconeogenesis		[36]

↑ higher; ↓ lower, ↔ similar.

background of Africans make them differentially susceptible to T2D compared to Europeans. Although there are data to suggest that hyperinsulinaemia may drive insulin resistance and hyperglycaemia in populations of European descent, to our knowledge, there are no prospective studies that have explored this possibility in populations of African descent. An understanding of the determinants of insulin sensitivity, secretion and clearance are essential to our understanding of the pathophysiology of T2D in African populations.

Determinants of insulin sensitivity, secretion and clearance

As the conventional paradigm has been that insulin resistance is the primary defect, resulting in a compensatory hyperinsulinaemia due to a combination of increased insulin secretion and reduced clearance, the majority of studies that have explored the pathophysiology of T2D in African populations have focused on the determinants of insulin resistance, rather than hyperinsulinaemia.

Obesity and body fat distribution

Within South Africa, 87% of T2D was attributable to an elevated BMI, with the attributable fraction being higher in women (94%) than men (75%) and highest in the 30–44-year age group [37]. This is related to the higher prevalence of obesity in women compared to men, with as many as 65% of women being classified as overweight or obese, compared to only 31% of men [23,25]. Notably, black African women are most affected by obesity in South Africa, with a prevalence as high as 40%

[23]. It was previously thought that obese black African women were 'metabolically healthy' and not prone to hyperglycaemia [38–40]. This may be related to their 'favourable' body fat distribution. Indeed, we have shown that for the same level of body fatness, measured using dual energy X-ray absorptiometry (DXA), black women had less central (trunk) fat mass than white South African women [41,42]. Further, we showed that for the same waist circumference, black South African women had less visceral adipose tissue (VAT) than white South African women [42–44] and that the lower VAT levels were similar to those of African Americans and Africans living in the USA [44] (Table 2). These ethnic differences were more pronounced in the obese women [42–44]. Centralization of body fat, in particular accumulation of VAT has been associated with insulin resistance and increased risk for T2D [45,46]. Conversely, lower-body peripheral (gluteo-femoral) fat deposition has been linked to improved insulin sensitivity [47]. This presents a paradox, as a consistent finding is lower VAT in combination with more abdominal and gluteo-femoral subcutaneous adipose tissue (SAT) in black Africans versus their white counterparts (Table 2), associated with lower levels of insulin sensitivity [12,42–43,48–51]. The reason for the ethnic disparities in body fat distribution is not known but differences in expression of oestrogen receptors (ER) in adipose tissue may partly explain these differences. Although ER expression in abdominal SAT did not differ by ethnicity, black African women had higher gluteal ER α and lower ER β expression than their white counterparts, which correlated with lower central fat mass and greater gynoid fat mass [52]. Further, we showed that lower accumulation of VAT relative to SAT in

Table 2. Ethnic differences in body fat distribution and ectopic fat deposition

Compared to white Europeans, black Africans present with		References
↓ central (trunk) fat mass		[41,42]
↓ visceral adipose tissue		[42–44]
↑ gluteo-femoral fat		[42,43]
↓ hepatic fat		[35–36,49–51,68,69]
↔↓ intramyocellular fat		[35,49,65,66]
↔↓ pancreatic fat		[49,75,76]

↑ higher; ↓ lower, ↔ similar.

black compared to white South African women may be related to their lower glucocorticoid receptor alpha expression in abdominal and gluteal SAT depots [53].

Majority of studies reporting ethnic differences in body fat distribution have relied on cross-sectional designs. More recently, in two longitudinal studies including free-living black South African women, we showed changes in body fat distribution with increasing age and adiposity [54,55]. Amongst premenopausal black South African women, we found that an increase in body weight (~9%) was associated with a relative increase in central fat mass (trunk, android, VAT and SAT), in particular an increase in VAT relative to abdominal SAT, and a concomitant decrease in peripheral fat mass (leg and gynoid) [54]. This change in body composition was associated with a decrease in insulin sensitivity and an increase in insulin response at follow-up [54]. Similarly, in a separate sample of middle-aged black South African women with normal glucose tolerance (NGT), an increase in body weight was associated with a greater increase in central vs. peripheral fat mass and a 2.5-fold greater increase in VAT than abdominal SAT [55]. Notably, baseline trunk fat mass and VAT, as well as the change in VAT were associated with greater odds of developing T2D, whereas baseline leg fat mass was associated with reduced T2D risk at follow-up [55]. Accordingly, we hypothesize that an age-related change in body composition in black African women, in particular VAT accumulation, may be a key element in the increased risk for developing T2D in black African women.

This background is of relevance for recent genetic studies [56,57], including those highlighted in the review by Yaghoobtar et al. in this symposium series (Yaghoobtar et al., in this issue), that provide empirical evidence to show that lower capacity for peripheral SAT accumulation is associated with insulin resistance and T2D [56,57]. Importantly, genetic studies are needed in populations of African ancestry to gain a full understanding of the pathogenic role of body fat distribution in insulin resistance and T2D prevalence. Nonetheless, we suggest that the redistribution of fat from the periphery to the central depots may be explained, in part, by the expression of adipogenic and lipogenic genes, which have been shown to be downregulated with obesity in the gluteal depot of black, but not white South African women [58]. Further, the gluteal SAT depot of obese black South African women was also

characterized as having a greater proportion of large adipocytes [59], as well as higher expression of proinflammatory genes [60] and genes related to fibrosis [61] compared to white South African women. This down-regulation of adipogenic and lipogenic genes leads to adipocyte hypertrophy and reduced capacity to store excess fatty acids in the peripheral depots. Adipocyte hypertrophy is associated with adipocyte hypoxia and fibrosis, macrophage recruitment and the release of inflammatory cytokines. These biological actions are associated with reduced insulin sensitivity in African women, as reviewed in greater detail previously [62].

Ectopic fat deposition

An inability to store excess lipids in peripheral SAT depots may lead to these lipids being stored in nonadipose tissue depots, namely the liver, skeletal muscle and pancreas. This ectopic fat deposition has been proposed to be closely associated with reduced insulin sensitivity and integral to the development of T2D [63,64]. Despite lower insulin sensitivity and greater risk for T2D, black Africans have less ectopic fat than their European counterparts, which is associated with their preferential peripheral SAT accretion that may act as a reservoir for excess lipids (Table 2). In terms of skeletal muscle, studies from South Africa, as well as the USA and UK have shown similar [35,49,65] or lower [66] intramyocellular (IMCL) levels in black compared to white women. In South African women, we showed a significant negative association between IMCL and insulin sensitivity in obese black African, but not white women [35]. This contrasts to studies in the USA [65,66] and UK [48,49] that showed a negative association between IMCL and insulin sensitivity in white European, but not black African women. However, the oxidative capacity and accumulation of lipid by-products, such as diacylglycerols and ceramides, are key determinants of insulin sensitivity. Indeed, DeLany et al. [67] showed that differences in peripheral insulin sensitivity between nonobese African American and European American women were related to a lower skeletal muscle mitochondrial oxidative capacity in the African American women. However, to our knowledge, there are no studies to date that have reported ethnic differences in skeletal muscle lipid metabolites.

We and others have shown that black African women have lower hepatic fat deposition compared to their white counterparts [35-36,49-51,68,69].

Whilst the reasons for this are not known, lower hepatic fat is associated with the lower VAT in black Africans [69]. Further, we have recently shown lower serum cholesterol ester stearyl-CoA desaturase-1 (SCD1) activity, a marker of hepatic de novo lipogenesis [70], in black compared to South African women (Goedecke et al, unpublished), data supported by a recent study from the USA [36]. Notably, we showed that the lower SCD-1 activity in black women was associated with their lower hepatic fat content, and both lower SCD-1 activity and hepatic fat were associated with higher hepatic insulin sensitivity in obese black African, but not obese white women [35] (Goedecke et al, unpublished). This suggests an increased sensitivity to ectopic fat deposition in black Africans [36,69]. In contrast, studies comparing west African and white European men with NGT and early T2D showed that hepatic fat was associated with hepatic insulin sensitivity in white European men, but not black west African men [49,50]. Accordingly, Goff et al. have challenged the existing theory that places ectopic fat deposition at the centre of the pathophysiology of T2D in men of black West African–Caribbean ethnicity [34].

Hepatic fat accumulation is considered the hallmark of reduced hepatic insulin clearance [71–73], but black Africans present with a phenotype of low hepatic fat in the face of low hepatic insulin clearance. When examining the relationship between hepatic fat and hepatic insulin clearance, no association was found in black African women living in the USA [12] or in premenopausal South African women (Fortuin-de Smidt, unpublished). In contrast, Hakim et al. [50] showed a trend for an association between hepatic fat and hepatic insulin clearance in west African men with early T2D. Further studies are required to dissect these relationships, as reduced hepatic insulin clearance has recently been hypothesized by Bergman et al. [74] to be the primary cause of T2D in at-risk individuals, including black Africans.

In contrast, Roy Taylor proposed that excess fat accumulation in the liver and pancreas are the prime drivers of abnormalities in insulin sensitivity and secretion that underlie T2D [73]. There are a dearth of studies that have explored the ethnic differences in pancreatic fat and the relationship between pancreatic fat and beta-cell function. Studies in UK have however shown an inverse relationship between pancreatic fat and insulin secretion in white European men with early T2D,

but not black west African men [75]. Preliminary data from premenopausal black South African women also did not show any association between pancreatic fat and insulin secretion (Fortuin-de Smidt, unpublished). In contrast, Szczepaniak et al. [76] showed a strong positive association between pancreatic fat and first phase insulin response during a FSIGT in nondiabetic African Americans, whilst European Americans with higher levels of pancreatic fat had a flatter positive relationship with insulin secretion. The positive association between pancreatic fat and insulin response in the nondiabetic population may be explained by the fact that free fatty acids stimulate insulin secretion, whereas in the diabetic state, the effects of lipotoxicity may override the stimulatory effects of the free fatty acids [76].

In summary, black Africans have less ectopic fat than their European counterparts despite lower insulin sensitivity and hyperinsulinaemia (Tables 1 and 2). This suggests marked ethnic differences in the pathogenesis of T2D. However, there is controversy as to whether black Africans are more sensitive to the effects of ectopic fat deposition than their European counterparts, or alternatively, whether ectopic fat is an innocent bystander and not involved in the pathogenesis of T2D in populations of African descent. Further, to our knowledge, it is not known whether the centralization of body fat with increasing age and adiposity in black Africans is associated with an increase in ectopic fat deposition, and whether this infers a greater risk for T2D. These conundrums can only be resolved by performing longitudinal studies, as well as intervention studies that perturb the system, including dietary and exercise intervention studies.

Sociodemographic and lifestyle factors

Within sub-Saharan Africa, rapid demographic (growing and ageing population), sociocultural (changes in lifestyle, activity patterns and eating habits), and economic (higher income, urbanization, changing food availability and evolving lifestyle and work practices) transitions are driving increases in risk factors and the prevalence of T2D [77]. It is therefore impossible to focus purely on physiological factors, without considering ethnic differences in sociocultural and lifestyle factors that impact on T2D risk. Indeed, a recent review of dietary surveys in South Africa showed that compared to other ethnicities, black South Africans

had the lowest mean dietary diversity scores, with diets very high in cereals (maize), bread and added sugar, and low in fruit and vegetables [78]. We hypothesize that overconsumption of cheaper, subsidized staple foods (maize and bread) with added sugar and fat that have high energy content and low nutritional value will not only lead to obesity (especially in the context of hyperinsulinaemia (Gower in this issue)), but also increase the risk of T2D via the independent effects on hyperinsulinaemia and insulin sensitivity. Indeed, overconsumption of energy, in particular carbohydrates, may increase hepatic fat and consequently reduce hepatic insulin clearance, leading to hyperinsulinaemia, which may have downstream effects on insulin sensitivity and consequently T2D risk [72]. However, this hypothesis is yet to be confirmed, particularly in an African population. To our knowledge, there are no studies that have examined the effects of a dietary intervention on insulin sensitivity, secretion or clearance in black African population. This is a priority as this information is key to our understanding of the pathogenesis of T2D and consequently essential for prevention and management of T2D in African populations.

A contributor to the increased T2D risk in South Africa is the low level of moderate-vigorous activity, on a population level. A surveillance study including 22 African countries reported that 84% of men and 76% of women met the World Health Organization (WHO) physical activity recommendations of achieving at least 150 min of moderate-vigorous activity or equivalent per week, according to the validated Global Physical Activity Questionnaire (GPAQ) [79]. Studies from South Africa using the same method showed that ~60% of South Africans meet these physical activity guidelines [80,81]. Most of this activity was accumulated through transport and work-related activities, with very low levels of leisure activity, which is typically undertaken at a higher intensity [80–82]. However, when physical activity was measured objectively using accelerometry in a sample of premenopausal black South African women, we showed that although 51% met the physical activity guidelines, the majority (59.4%) of the day was spent in sedentary behaviours. Only 3.5% of the day was spent performing moderate-to-vigorous activities, whereas 37% of the day was spent in light activity, with consequently very low cardiorespiratory fitness ($\text{VO}_{2\text{peak}} \pm 25 \text{ mL kg}^{-1} \text{ min}^{-1}$) [83]. Both cardiorespiratory fitness and light

physical activity, and not moderate-to-vigorous activity, were associated with lower total body fat, VAT and insulin sensitivity [83]. Based on these results, we completed a 12-week exercise intervention, consisting of both aerobic and resistance exercise, in order to identify mechanisms underlying changes in insulin sensitivity and hyperinsulinaemia in obese black South African women [84]. Exercise training improved cardiorespiratory fitness and resulted in a small, but significant decrease in BMI and gynoid fat mass [85]. The 12-week exercise intervention improved insulin sensitivity without a concomitant change in insulin secretion, clearance or beta-cell function (Fortuin de-Smidt et al., unpublished). These findings suggest that either a longer duration training intervention, or more likely, exercise training in combination with dietary intervention is required to alter both insulin sensitivity and hyperinsulinaemia in black African populations.

Finally, important barriers to reduce T2D risk are differences in perceptions of body size and body image [86]. Indeed, black Africans have a greater body size tolerance than their white counterparts [87], and regard body fatness as a sign of health, beauty, respect, happiness, fertility and not having HIV (as reviewed previously [62,88]). Interestingly, we recorded changes in the perception of body size in the obese black South African women in response to the 12-week exercise intervention. At baseline, although obesity was part of the inclusion criteria, only 5% of the women considered themselves obese, whereas 20% considered themselves to be of normal body weight. At follow-up, although the mean weight loss over the 12-week exercise intervention was only ~1 kg and the average post-training BMI was 33.8 kg m^{-2} , no women considered themselves to be obese and 40% considered themselves to be normal weight (Phiri et al., unpublished). These findings underscore the complexities surrounding perceptions regarding body size, weight loss and weight-loss interventions, and, together with the differences in socioeconomic status, cultural ideals and community perceptions [86,87], will impact future lifestyle interventions in black African populations.

Conclusions

Black Africans, irrespective of their geographical location, present with a phenotype of low insulin sensitivity and hyperinsulinaemia related to

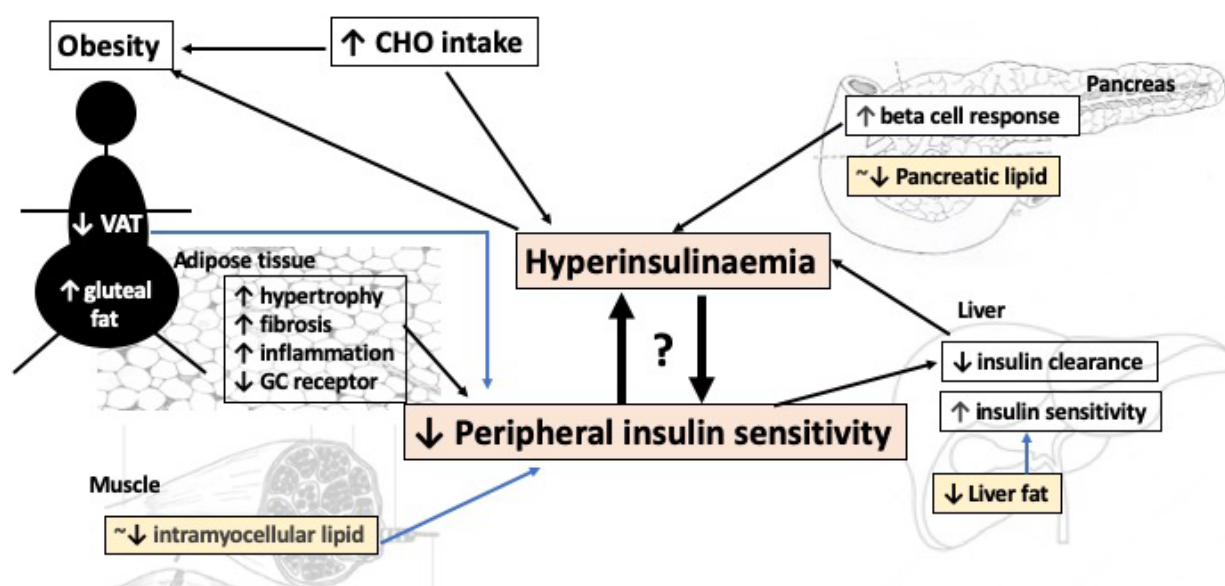


Fig. 2 Schematic representation of the pathogenesis of type 2 diabetes in African women. Black Africans present with a phenotype of low insulin sensitivity and hyperinsulinaemia as a result of increased insulin secretion and reduced hepatic insulin clearance. Whether hyperinsulinaemia precedes insulin resistance or is merely a compensatory mechanism is yet to be determined. Paradoxically, black Africans have lower visceral adipose tissue and ectopic fat deposition and greater peripheral (gluteo-femoral) fat deposition than their European counterparts. It is not known whether black Africans are more sensitive to the effects of ectopic fat deposition than their white Europeans, or whether ectopic fat is not an important mediator of T2D in black Africans. Arrows indicate comparisons to white women of European descent: ↑ higher, ↓ lower, ~ similar to white European women. Solid line indicates a positive association. Blue arrows suggest increased sensitivity to the effects of ectopic fat on insulin sensitivity.

increased insulin secretion and reduced hepatic insulin clearance (Fig. 2). Hyperinsulinaemia may be the primary defect that explains the higher T2D prevalence in black African compared to European populations. Dietary interventions and genetic studies are required to address this hypothesis. The outcome of these studies will help us to gain a better understanding of the pathophysiology of T2D with the ultimate aim to suggest effective prevention and management solutions.

Funding

Funding by the South African National Research Foundation (NRF) and the Swedish Foundation for International Cooperation in Research and Higher Education (STINT) Sweden/South Africa Research Collaboration Programme (UID: 106784), the South African Medical Research Council, Swedish Heart and Lung Foundation, the Novo Nordisk Foundation and Stellenbosch Institute for Advanced Study.

Conflict of interest

The authors have no conflicts of interest to declare.

Author contribution

Julia H Goedecke: Conceptualization (lead); Data curation (lead); Funding acquisition (supporting); Investigation (lead); Project administration (lead); Writing-original draft (lead); Writing-review & editing (equal). Tommy Olsson: Conceptualization (supporting); Data curation (supporting); Funding acquisition (lead); Investigation (supporting); Project administration (supporting); Writing-original draft (supporting); Writing-review & editing (equal).

References

- 1 International Diabetes Federation. *IDF Diabetes Atlas*, 8th edn. Belgium: International Diabetes Federation, 2017: 1–25.
- 2 Statistics South Africa. Mortality and causes of death in South Africa, 2016: Findings from death notification. *Statistics South Africa* 2016; **2018**: 1–142.

- 3 Peer N, Steyn K, Lombard C, Lambert EV, Vythilingum B, Rising Levitt NS. Diabetes prevalence among urban-dwelling black south Africans. Herder C, ed. *PLoS ONE* 2012; **7**: e43336.
- 4 Kahn SE, Prigeon RL, McCulloch DK *et al.* Quantification of the relationship between insulin sensitivity and beta-cell function in human subjects. Evidence for a hyperbolic function. *Diabetes* 1993; **42**: 1663–72.
- 5 Goedecke JH, Dave JA, Faulenbach MV *et al.* Insulin response in relation to insulin sensitivity: an appropriate beta-cell response in black South African women. *Diabetes Care* 2009; **32**: 860–5.
- 6 van der Merwe MT, Crowther NJ, Schlaphoff GP, Gray IP, Joffe BI, Lönnroth PN. Evidence for insulin resistance in black women from South Africa. *Int J Obes Relat Metab Disord* 2000; **24**: 1340–6.
- 7 Osei K, Schuster DP. Ethnic differences in secretion, sensitivity, and hepatic extraction of insulin in black and white Americans. *Diabet Med* 1994; **11**: 755–62.
- 8 Kodama K, Tojjar D, Yamada S, Toda K, Patel CJ, Butte AJ. Ethnic differences in the relationship between insulin sensitivity and insulin response: a systematic review and meta-analysis. *Diabetes Care* 2013; **36**: 1789–96.
- 9 Lorenzo C, Wagenknecht LE, Rewers MJ *et al.* Disposition index, glucose effectiveness, and conversion to type 2 diabetes: the insulin resistance atherosclerosis study (IRAS). *Diabetes Care* 2010; **33**: 2098–103.
- 10 Chandler-Laney PC, Phadke RP, Granger WM *et al.* Age-related changes in insulin sensitivity and β -cell function among European-American and African-American women. *Obesity* 2011; **19**: 528–35.
- 11 Chandler-Laney PC, Phadke RP, Granger WM *et al.* Adiposity and β -cell function: relationships differ with ethnicity and age. *Obesity* 2010; **18**: 2086–92.
- 12 Chung ST, Galvan-De La Cruz M, Aldana PC *et al.* Postprandial insulin response and clearance among black and white women: the federal women's study. *J Clin Endocrinol Metab* 2018; **104**: 181–92.
- 13 Writing group members, Lloyd-Jones D, Adams RJ *et al.* Disease and stroke statistics–2010 update: a report from the American heart association. *Circulation* 2010; **121**: e46–e215.
- 14 Ferrannini E, Wahren J, Faber OK, Felig P, Binder C, DeFronzo RA. Splanchnic and renal metabolism of insulin in human subjects: a dose-response study. *Am J Physiol* 1983; **244**: E517–27.
- 15 Meier JJ, Veldhuis JD, Butler PC. Pulsatile insulin secretion dictates systemic insulin delivery by regulating hepatic insulin extraction in humans. *Diabetes* 2005; **54**: 1649–56.
- 16 Osei K, Schuster DP, Owusu SK, Amoah AG. Race and ethnicity determine serum insulin and C-peptide concentrations and hepatic insulin extraction and insulin clearance: comparative studies of three populations of West African ancestry and white Americans. *Metab Clin Exp* 1997; **46**: 53–8.
- 17 Piccinini F, Polidori DC, Gower BA, Fernández JR, Bergman RN. Dissection of hepatic versus extra-hepatic insulin clearance: Ethnic differences in childhood. *Diabet Obesit Metabol* 2018; **54**: 1649–7.
- 18 Mittelman SD, Van Citters GW, Kim SP *et al.* Longitudinal compensation for fat-induced insulin resistance includes reduced insulin clearance and enhanced beta-cell response. *Diabetes* 2000; **49**: 2116–25.
- 19 Galderisi A, Polidori D, Weiss R *et al.* Lower Insulin Clearance parallels a reduced insulin sensitivity in obese youths and is associated with a decline in β -cell function over time. *Diabetes* 2019; **68**: 2074–84.
- 20 Lee CC, Haffner SM, Wagenknecht LE *et al.* Insulin clearance and the incidence of Type 2 diabetes in hispanics and African Americans: the IRAS family study. *Diabetes Care* 2013; **36**: 901–7.
- 21 Goedecke JH, George C, Veras K *et al.* Sex differences in insulin sensitivity and insulin response with increasing age in black South African men and women. *Diabetes Res Clin Pract* 2016; **122**: 207–14.
- 22 Geer EB, Shen W. Gender differences in insulin resistance, body composition, and energy balance. *Gender Med* 2009; **6**: 60–75.
- 23 Shisana O, Labadarios D, Rehle T *et al.* South African National Health and Nutrition Examination Survey (SANHANES-1). *HSRC Press* 2013; 1–423.
- 24 Nienaber-Rousseau C, Sotunde OF, Ukegbu PO *et al.* Socio-demographic and lifestyle factors predict 5-year changes in adiposity among a group of black south African Adults. *IJERPH* 2017; **14**: 1089.
- 25 Ng M, Fleming T, Robinson M *et al.* Global, regional, and national prevalence of overweight and obesity in children and adults during 1980–2013: a systematic analysis for the Global Burden of Disease Study 2013. *The Lancet* 2014; **384**: 766–81.
- 26 Motala AA, Esterhuizen T, Gouws E, Pirie FJ, Omar MAK. Diabetes and other disorders of glycemia in a rural South African community: prevalence and associated risk factors. *Diabetes Care* 2008; **31**: 1783–8.
- 27 Mohandas C, Bonadonna R, Shojee-Moradie F *et al.* Ethnic differences in insulin secretory function between black African and white European men with early type 2 diabetes. *Diabetes, Obes Metab* 2018; **20**: 1678–87.
- 28 Berman P, Collins M, Baumgarten I *et al.* Association between the 4 bp proinsulin gene insertion polymorphism (IVS-69) and body composition in black South African women. *Obesity (Silver Spring)* 2009; **17**: 1298–300.
- 29 Rizza RA, Mandarino LJ, Genest J, Baker BA, Gerich JE. Production of insulin resistance by hyperinsulinaemia in man. *Diabetologia* 1985; **28**: 70–5.
- 30 Shanik MH, Xu Y, Skrha J, Dankner R, Zick Y, Roth J. Insulin resistance and hyperinsulinemia: is hyperinsulinemia the cart or the horse? *Diabetes Care* 2008; **31**: S262–8.
- 31 Del Prato S, Leonetti F, Simonson DC, Sheehan P, Matsuda M, DeFronzo RA. Effect of sustained physiologic hyperinsulinaemia and hyperglycaemia on insulin secretion and insulin sensitivity in man. *Diabetologia* 1994; **37**: 1025–35.
- 32 Martin C, Desai KS, Steiner G. Receptor and postreceptor insulin resistance induced by in vivo hyperinsulinemia. *Can J Physiol Pharmacol* 1983; **61**: 802–7.
- 33 Tricò D, Natali A, Arslanian S, Mari A, Ferrannini E. Identification, pathophysiology, and clinical implications of primary insulin hypersecretion in nondiabetic adults and adolescents. *JCI Insight* 2018; **3**: E53.
- 34 Goff LM, Ladwa M, Hakim O, Bello O. Ethnic distinctions in the pathophysiology of type 2 diabetes: a focus on black African-Caribbean populations. *Proc Nutr Soc* 2019; **3**: 1–10.

- 35 Goedecke JH, Keswell D, Weinreich C *et al.* Ethnic differences in hepatic and systemic insulin sensitivity and their associated determinants in obese black and white South African women. *Diabetologia* 2015; **58**: 2647–52.
- 36 Chung ST, Courville AB, Onuzuruike AU *et al.* Gluconeogenesis and risk for fasting hyperglycemia in Black and White women. *JCI Insight* 2018; **3**: e67–14.
- 37 Joubert J, Norman R, Bradshaw D *et al.* Estimating the burden of disease attributable to excess body weight in South Africa in 2000. *S Afr Med J* 2007; **97**: 683–90.
- 38 Walker A, Adam F, Walker BF. World pandemic of obesity. *Public Health* 2001; **115**: 368–72.
- 39 Walker AR, Segal I. The puzzle of obesity in the African Black female. *The Lancet* 1980; **1**: 263.
- 40 Walker AR, Walker BF, Manetsi B, Tsotetsi NG, Walker AJ. Obesity in black women in Soweto, South Africa: minimal effects on hypertension, hyperlipidaemia and hyperglycaemia. *J R Soc Health* 1990; **110**: 101–3.
- 41 Rush EC, Goedecke JH, Jennings C *et al.* BMI, fat and muscle differences in urban women of five ethnicities from two countries. *Int J Obes Relat Metab Disord* 2007; **31**: 1232–9.
- 42 Keswell D, Tootla M, Goedecke JH. Associations between body fat distribution, insulin resistance and dyslipidaemia in black and white South African women. *Cardiovasc J Afr* 2016; **27**: 1–7.
- 43 Goedecke JH, Levitt NS, Lambert EV *et al.* Differential effects of abdominal adipose tissue distribution on insulin sensitivity in black and white South African women. *Obesity* 2009; **17**: 1506–12.
- 44 Sumner AE, Micklesfield LK, Ricks M *et al.* Waist circumference, BMI, and visceral adipose tissue in white women and women of african descent. *Obesity (Silver Spring)* 2010; **19**: 671–4.
- 45 Preis SR, Massaro JM, Robins SJ *et al.* Abdominal Subcutaneous and Visceral Adipose Tissue and Insulin Resistance in the Framingham Heart Study. *Obesity* 2009; **18**: 2191–8.
- 46 Després JP. Abdominal obesity as important component of insulin-resistance syndrome. *Nutrition* 1993; **9**: 452–9.
- 47 Manolopoulos KN, Karpe F, Frayn KN. Gluteofemoral body fat as a determinant of metabolic health. *Int J Obes Relat Metab Disord* 2010; **34**: 949–59.
- 48 Bello O, Mohandas C, Shojee-Moradie F *et al.* Black African men with early type 2 diabetes have similar muscle, liver and adipose tissue insulin sensitivity to white European men despite lower visceral fat. *Diabetologia* 2019; **62**: 835–44.
- 49 Hakim O, Bello O, Ladwa M *et al.* Ethnic differences in hepatic, pancreatic, muscular and visceral fat deposition in healthy men of white European and black west African ethnicity. *Diabetes Res Clin Pract* 2019; **156**: 107866.
- 50 Hakim O, Bello O, Bonadonna RC *et al.* Ethnic differences in intrahepatic lipid and its association with hepatic insulin sensitivity and insulin clearance between men of Black and White ethnicity with early type 2 diabetes. *Diabetes Obes and Metab* 2019; **21**: 2163–8.
- 51 Nazare JA, Smith JD, Borel AL *et al.* Ethnic influences on the relations between abdominal subcutaneous and visceral adiposity, liver fat, and cardiometabolic risk profile: the International Study of Prediction of Intra-Abdominal Adiposity and Its Relationship With Cardiometabolic Risk/ Intra-Abdominal Adiposity. *Am J Clin Nutrition* 2012; **96**: 714–26.
- 52 Goedecke JH, Tootla M, Keswell D. Ethnic differences in regional adipose tissue oestrogen receptor gene expression. *Endocrine Connections* 2019; **8**: 32–8.
- 53 Goedecke JH, Chorell E, Livingstone DEW *et al.* Glucocorticoid receptor gene expression in adipose tissue and associated metabolic risk in black and white South African women. *Int J Obes* 2015; **39**: 303–11.
- 54 Chantler S, Dickie K, Micklesfield LK, Goedecke JH. Longitudinal changes in body fat and its distribution in relation to cardiometabolic risk in black South African women. *Metab Syndr Relat Disord* 2015; **13**: 381–8.
- 55 Mtintsilana A, Micklesfield LK, Chorell E, Olsson T, Goedecke JH. Fat redistribution and accumulation of visceral adipose tissue predicts type 2 diabetes risk in middle-aged black South African women: a 13-year longitudinal study. *Nutr Diabetes* 2019; **9**: 12.
- 56 Ji Y, Yiorakas AM, Frau F *et al.* Genome-wide and abdominal mri data provide evidence that a genetically determined favorable adiposity phenotype is characterized by lower ectopic liver fat and lower risk of type 2 diabetes, heart disease, and hypertension. *Diabetes* 2019; **68**: 207–19.
- 57 Lotta LA, Gulati P, Day FR *et al.* Integrative genomic analysis implicates limited peripheral adipose storage capacity in the pathogenesis of human insulin resistance. *Nat Genet* 2017; **49**: 17–26.
- 58 Goedecke JH, Evans J, Keswell D *et al.* Reduced gluteal expression of adipogenic and lipogenic genes in black south African women is associated with obesity-related insulin resistance. *J Clin Endocrinol Metabolism* 2011; **96**: E2029–33.
- 59 Goedecke JH, Levitt NS, Evans J *et al.* The role of adipose tissue in insulin resistance in women of African ancestry. *J Obes* 2013; **2013**: 952916.
- 60 Evans J, Goedecke JH, Söderström I *et al.* Depot- and ethnic-specific differences in the relationship between adipose tissue inflammation and insulin sensitivity. *Clin Endocrinol (Oxf)* 2010; **74**: 51–9.
- 61 Kotzé-Hörstmann LM, Keswell D, Adams K, Dlamini T, Goedecke JH. Hypoxia and extra-cellular matrix gene expression in adipose tissue associates with reduced insulin sensitivity in black South African women. Liske M, Kotzé-Hörstmann, Dheshnie Keswell, Kevin Adams, Thandiwe Dlamini & Julia H. Goedecke. *Endocrine* 2017; **55**: 144–52.
- 62 Goedecke JH, Mtintsilana A, Dlamini SN, Kengne AP. Type 2 diabetes mellitus in African women. *Diabetes Res Clin Pract* 2017; **123**: 87–96.
- 63 Shulman GI. Ectopic fat in insulin resistance, dyslipidemia, and cardiometabolic disease. *N Engl J Med* 2014; **371**: 1131–41.
- 64 Sattar N, Gill JM. Type 2 diabetes as a disease of ectopic fat? *BMC Med* 2014; **12**: 123.
- 65 Ingram KH, Lara-Castro C, Gower BA *et al.* Intramyocellular lipid and insulin resistance: differential relationships in European and African Americans. *Obesity* 2009; **19**: 1469–75.
- 66 Smith LM, Yao-Borengasser A, Starks T, Tripputi M, Kern PA, Rasouli N. Insulin resistance in African-American and caucasian women: differences in lipotoxicity, adipokines, and gene expression in adipose tissue and muscle. *J Clin Endocrinol Metab* 2010; **95**: 4441–8.
- 67 DeLany JP, Dubé JJ, Standley RA *et al.* Racial differences in peripheral insulin sensitivity and mitochondrial capacity in

- the absence of obesity. *J Clin Endocrinol Metab* 2014; **99**: 4307–14.
- 68 Naran NH, Haagenzen M, Crowther NJ. Steatosis in South African women: How much and why? Sonderup MW, ed. *PLoS ONE* 2018; **13**: e0191388.
 - 69 Guerrero R, Vega GL, Grundy SM, Browning JD. Ethnic differences in hepatic steatosis: an insulin resistance paradox? *Hepatology* 2009; **49**: 791–801.
 - 70 Peter A, Cegan A, Wagner S *et al*. Hepatic lipid composition and stearoyl-coenzyme A desaturase 1 mRNA expression can be estimated from plasma VLDL fatty acid ratios. *Clin Chem* 2009; **55**: 2113–20.
 - 71 Kotronen A, Juurinen L, Tiikkainen M, Vehkavaara S, Yki-Jarvinen H. Increased liver fat, impaired insulin clearance, and hepatic and adipose tissue insulin resistance in type 2 diabetes. *Gastroenterology* 2008; **135**: 122–30.
 - 72 Bojsen-Møller KN, Lundsgaard A-M, Madsbad S, Kiens B, Holst JJ. Hepatic insulin clearance in regulation of systemic insulin concentrations—role of carbohydrate and energy availability. *Diabetes* 2018; **67**: 2129–36.
 - 73 Taylor R. Pathogenesis of type 2 diabetes: tracing the reverse route from cure to cause. *Diabetologia* 2008; **51**: 1781–9.
 - 74 Bergman RN, Piccinini F, Kabir M, Kolka CM, Ader M. Hypothesis: role of reduced hepatic insulin clearance in the pathogenesis of type 2 diabetes. *Diabetes* 2019; **68**: 1709–16.
 - 75 Hakim O, Bonadonna RC, Mohandas C *et al*. Associations between pancreatic lipids and β -cell function in black African and white European men with type 2 diabetes. *J Clin Endocrinol Metabolism* 2018; **104**: 1201–10.
 - 76 Szczepaniak LS, Victor RG, Mathur R *et al*. Pancreatic steatosis and its relationship to β -cell dysfunction in humans: racial and ethnic variations. *Diabetes Care* 2012; **35**: 2377–83.
 - 77 Atun R, Davies JI, Gale EAM *et al*. Diabetes in sub-Saharan Africa: from clinical care to health policy. *Lancet Diabetes Endocrinol* 2017; **5**: 622–67.
 - 78 Mchiza ZJ, Steyn NP, Hill J *et al*. A Review of dietary surveys in the adult South African population from 2000 to 2015. *Nutrients* 2015; **7**: 8227–50.
 - 79 Guthold R, Louazani SA, Riley LM *et al*. Physical activity in 22 African countries. *Am J Prev Med* 2011; **41**: 52–60.
 - 80 Gradidge PJ-L, Crowther NJ, Chirwa ED, Norris SA, Micklesfield LK. Patterns, levels and correlates of self-reported physical activity in urban black Soweto women. *BMC Public Health* 2014; **14**: 934.
 - 81 Dickie K, Micklesfield LK, Chantler S, Lambert EV, Goedecke JH. Meeting physical activity guidelines is associated with reduced risk for cardiovascular disease in black South African women; a 5.5-year follow-up study. *BMC Public Health* 2014; **14**: 1–11.
 - 82 Goedecke JH, Utzschneider K, Faulenbach MV *et al*. Ethnic differences in serum lipoproteins and their determinants in South African women. *Metab Clin Exp* 2010; **59**: 1341–50.
 - 83 Dickie K, Micklesfield LK, Chantler S, Lambert EV, Goedecke JH. Cardiorespiratory fitness and light-intensity physical activity are independently associated with reduced cardiovascular disease risk in urban black south African Women: a cross-sectional study. *Metab Syndr Relat Disord* 2016; **14**: 23–32.
 - 84 Goedecke JH, Mendham AE, Clamp L *et al*. An Exercise Intervention to Unravel the Mechanisms Underlying Insulin Resistance in a Cohort of Black South African Women: Protocol for a Randomized Controlled Trial and Baseline Characteristics of Participants. *JMIR Res Protoc* 2018; **7**: e75–19.
 - 85 Clamp LD, Mendham AE, Kroff J, Goedecke JH. Higher baseline fat oxidation promotes gynoid fat mobilization in response to a 12 week exercise intervention in sedentary, obese black South African women. *Appl Physiol Nutr Metab* 2019; **45**: 327–35.
 - 86 Draper CE, Davidowitz KJ, Goedecke JH. Perceptions relating to body size, weight loss and weight-loss interventions in black South African women: a qualitative study. *Public Health Nutr* 2015; **19**: 548–56.
 - 87 Mchiza ZJ, Parker W-A, Makoe M, Sewpaul R, Kupamupindi T, Labadarios D. Body image and weight control in South Africans 15 years or older: SANHANES-1. *BMC Public Health* 2015; **15**: 992.
 - 88 Micklesfield LK, Lambert EV, Hume DJ *et al*. Sociocultural, environmental and behavioural determinants of obesity in black South African women. *Cardiovascular J Afr* 2013; **24**: 1–7.
- Correspondence:** Julia H. Goedecke, Non-Communicable Diseases Research Unit, South African Medical Research Council, PO Box 19070, Tygerberg 7505, South Africa.
(email: Julia.goedecke@mrc.ac.za)